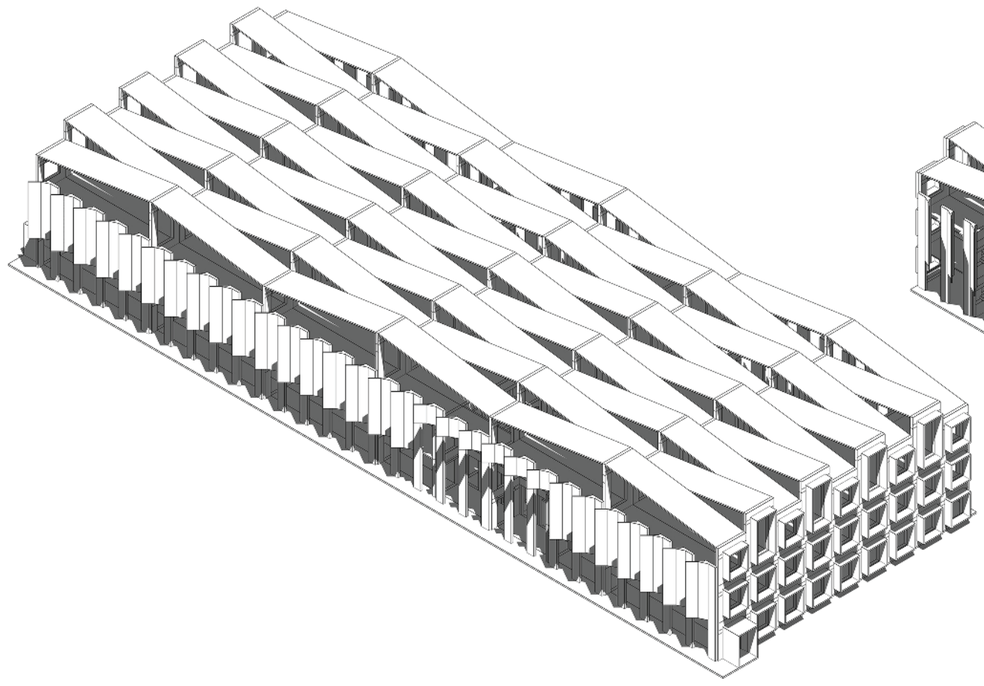
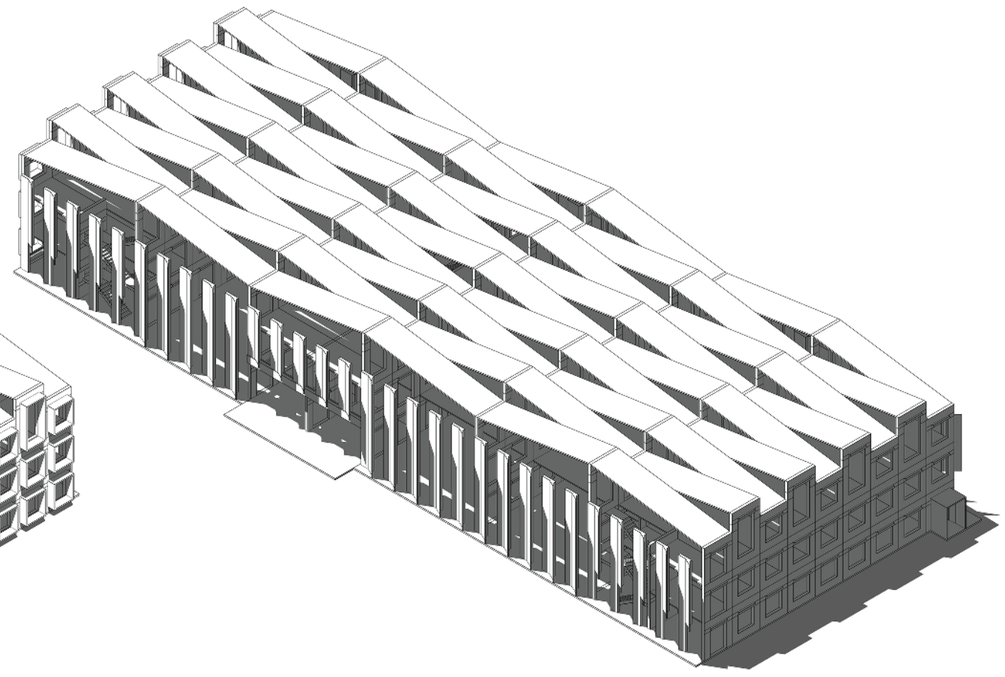




DELTA
TYPE

D.3
VARIANT

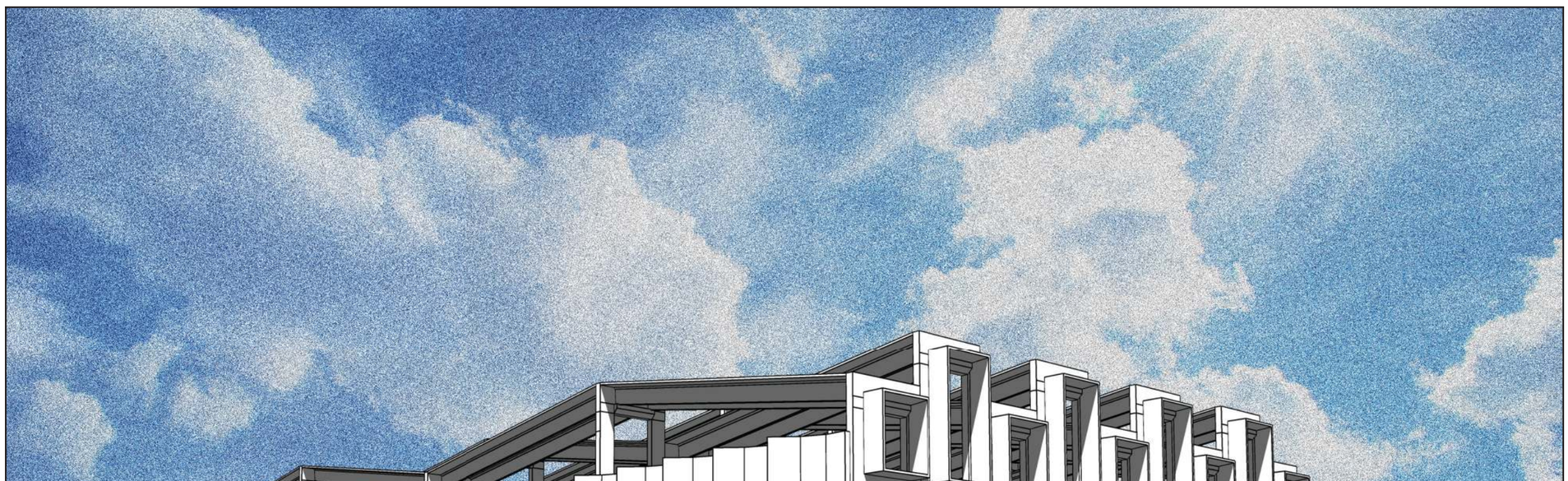
VIRTUES

Delta extends a common morphology found in parking garages, that of deep horizontal double-tees supported on structural lite walls. Creating 5000 square-foot spaces free of vertical support, this variant combines a practical and poetic definition of space. Selected double-tees are deleted in the floor structure to create double-height spaces and a connector bridge in the main lobby.

An expressive roof form rises and falls in alternating bays, creating clerestory apertures for dynamic daylight effect. This effect is revealed in section at the end bays. Interior space is characterized by a density of structure in the plane of lite walls, contrasted with expansive lateral space.

Enclosure on the long east and west sides is comprised of vertical insulated panels. They are segmented in profile and fan out from transparent glazing to softly modify daylight and inhibit solar gain. A deep porch is defined on the western main entry by deleting strategic panel segments.

Enclosure on the short north and south sides is comprised of structural insulated panels, expressing the internal load-bearing sequence of lite-walls. The south side has deep shadow hoods surrounding the glazed apertures, while the north side has glazing set flush with the face of panels.



MEASURES

Building Footprint
108 feet by 300 feet = 32,400 square feet

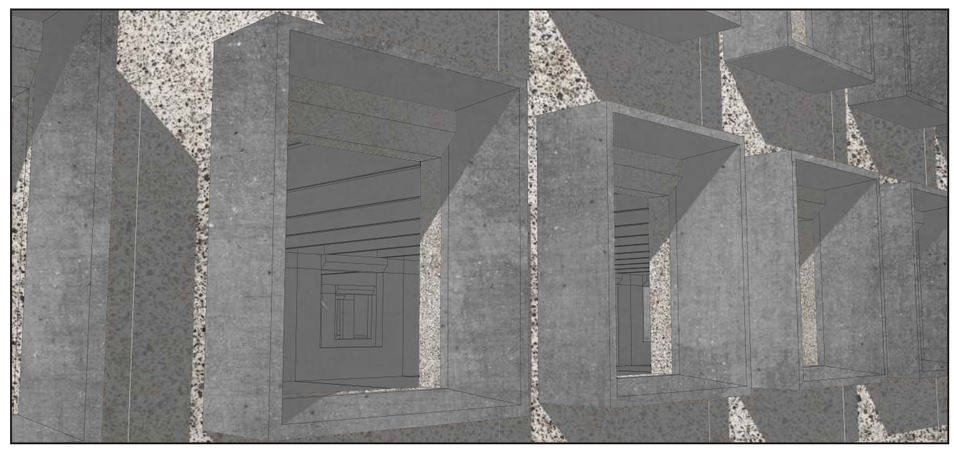
Structural Bay Size
12 feet by 50 feet
50 feet by 108 feet column-free space

Building Height
14 foot floor-to-floor height
Three stories at 42 feet to 49 feet total height

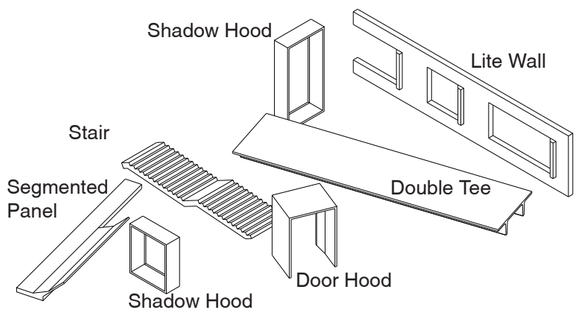
Area of Precast Elements
122,778 square feet of structure and enclosure

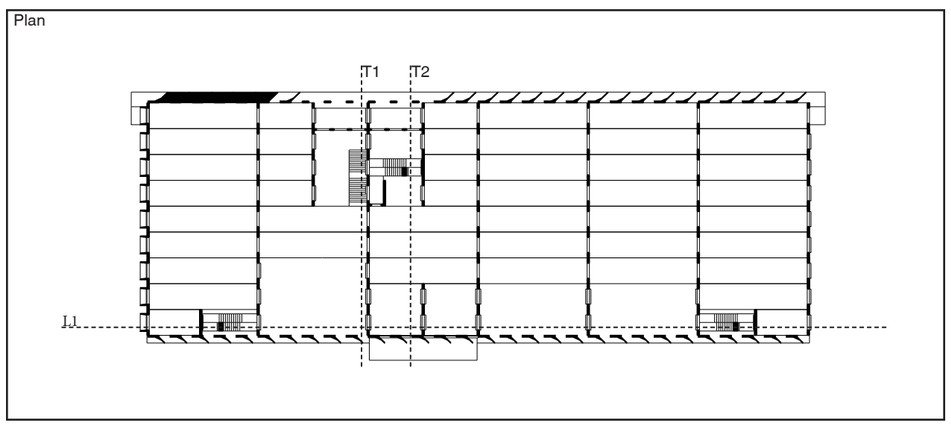
Enclosure Glazing Ratio
26% of total thermal enclosure

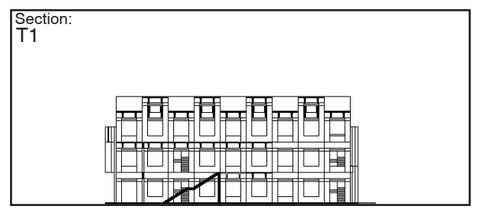


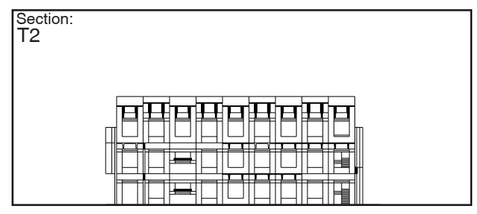


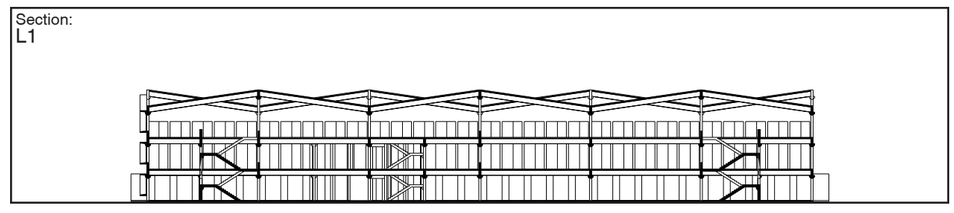
FINITE ELEMENTS











PRECAST
PROTOTYPES

